

WHEN RESAMPLING CHANGES PAIRED AUDIO-DEEPPAKE DETECTOR COMPARISONS

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ABSTRACT

We measure the sensitivity of paired equal-error-rate (EER) differences on fixed detector scores. We call a pair separated when its all-pair sensitivity band excludes zero; these bands carry no population-coverage claim. A 2.153-point EER advantage between two Speech DF Arena detectors loses simultaneous-band separation when speakers and attacks replace trials as the resampling unit. The same change occurs for XLSR-Mamba versus XLSR-Conformer, despite substantial cancellation from system pairing. Trial-to-joint separation falls from 26/28 to 18/28 on primary ASVspoof 2021 DF; a separate four-detector SpoofCeleb roster changes from 6/6 to 3/6. Speaker-only matches the primary joint indicators; attack-only does so on SpoofCeleb. Measured sensitivity concentrates in particular observed groups, while all SSL-versus-baseline comparisons survive the tested perturbations. Paired comparisons can therefore remain sensitive to group perturbations after substantial common variation cancels.

Index Terms— Audio deepfake detection, paired evaluation, resampling, EER

1. INTRODUCTION

Does pairing detectors on the same trials cancel the effect of perturbing speakers and attacks? For HuBERT-ECAPA versus WavLM-ECAPA in Speech DF Arena [1], a -2.153 -point EER gap has a trial-resampling band excluding zero, yet its joint speaker-attack band includes zero. XLSR-Mamba versus XLSR-Conformer shows the same change at -0.389 points (Fig. 1). In both comparisons the scores and point gap stay fixed. What changes is the unit whose multiplicities vary.

ASVspoof 2021 DF has 533,928 evaluation trials but only 93 speakers and 110 attacks. Detectors can move together when an observed group gains weight, reducing the variation of their difference. Dependence in individual scores therefore does not by itself establish sensitivity of a paired EER contrast. The measurement question is which comparisons change when the perturbation unit changes while threshold refitting, system pairing and the comparison family are held fixed.

We measure whether paired detector comparisons survive changes in the resampling unit and in observed-group weights. The primary roster contains four author-released

self-supervised (SSL) detectors and four organizer baselines; eleven Arena score files for the same trials form a separate roster. Named modern comparisons lose separation even after pairing cancels much of their marginal variation. One-factor perturbations and delete-one-group refits localize that sensitivity, and a single-source SpoofCeleb check gives a contrasting factor pattern. The SSL-versus-baseline boundary survives every tested resampling, fixed-weight and source-deletion check.

2. RELATED WORK

Benchmark reporting. ASVspoof 2017 reported EER intervals [2]; ASVspoof 2021 Fig. 4(c) used the Bengio–Mariéthoz test [3] with Holm correction [4]. Its shared-trial variant accounts for two systems using the same trials, not repeated speakers and attacks. We found no uncertainty analysis in the later journal overview [5], ASVspoof 5 [6], or Arena report [1]; a recent SSL benchmark reports point EERs [7]. Other work reports trial-bootstrap ranks [8], EER spreads [9], corrected pairwise tests [10], or fixed-threshold intervals [11]. Training-seed variation also moves rankings [12]; our scores hold training fixed.

Dependent paired comparison. User-aware biometric and speech resampling [13, 14], correlation-corrected paired tests [15], and synchronized resampling for correlated classifiers [16] precede this study. Our speaker and spoof-attack factors are crossed [17, 18, 19], distinct from the nested-observation designs studied in [20]. The contribution is a matched measurement of paired EER comparisons under these perturbations, with every threshold refitted.

Evaluation design. Aggregation weights [21], channel presentation [22], representations [23], controlled generator/speaker perturbations [24] and corpus overlap [25] affect deepfake evaluation. Identity perturbations [26], speaker-adversarial training [27] and speaker/synthesizer familiarity [28] address related sensitivities. Here the scores are fixed and evaluation weights vary. Simultaneous rank intervals are established [29]. Because selecting only neighbors in the observed point-EER ordering uses the outcomes, we include every detector pair.

3. METHOD

3.1. EER contrast and separation event

Two deterministic systems on a fixed trial list have no sampling uncertainty until a perturbation law is declared. Scores are oriented so larger means bona fide and sorted in ascending order. At each ordered position, bona-fide mass at or below it is the false-reject rate (FRR) and spoof mass above it is the false-accept rate (FAR); the first position minimizing $|FRR - FAR|$ is selected and EER is their mean, without interpolation. For ordered pair (A, B) , $\Delta EER_{A,B} = 100(EER_A - EER_B)$ percentage points (negative favors A); every replicate refits both thresholds. A pair is *separated under procedure h* when its simultaneous band excludes zero; write this indicator as $I_{p,h}$.

Across the eight systems and twelve fixed weighting rules, evaluating EER only at distinct-score boundaries changes each point EER by less than 0.001 percentage point. Re-computing distinct-score thresholds on all 45,000 checked weighted replicates preserved every separation indicator; the maximum band-endpoint change was 0.00024 point (supplement, S4d).

3.2. Perturbation laws and scope

The trial bootstrap independently resamples trials within class; for SpoofCeleb only, the archived plan samples the 91,130 trial indices from the pooled list with replacement, so class counts vary across replicates. In the speaker $\times$ attack product-weight (PW) bootstrap [19, 30], speakers and spoof attacks are independently resampled with replacement; a spoof trial receives the product of its speaker and attack multiplicities, while a bona-fide trial receives only its speaker multiplicity. Speaker-only fixes attack multiplicities at one; attack-only fixes speaker and bona-fide trial weights at one. Each of these four procedures uses 5,000 replicates. Each replicate’s weights are shared across systems; procedures have separate random streams (S4a).

Interpretation. The identified object is the change in ΔEER and $I_{p,h}$ on the fixed scores, not population uncertainty. These are fixed-score sensitivity bands, not population confidence intervals or confidence sets for ranks. Four VCC spoof strata contain only 4/4/4/6 speakers, so no 21DF separation result transfers to new speakers or attacks. Conservatism results for crossed-array means [30] do not establish EER coverage. The one-factor, Monte Carlo, mean/sign, pairing, tie and influence diagnostics (S4a–S4e) were designed after examining the primary trial-versus-joint results and are exploratory, not preregistered. Indicator agreement is not a decomposition of the effect or a causal attribution to speaker identity: observed groups bundle trials, attacks and recording provenance, not intrinsic voice properties. Different detector rosters prevent attributing the primary/SpoofCeleb factor contrast to dataset alone. A band containing zero means the

procedure does not separate that pair; it does not establish equality. Passing both checks does not guarantee an ordering under every possible reweighting or select a deployment ranking. Population confidence requires a defensible acquisition model and an inferential procedure valid under its dependence structure.

3.3. All-pair simultaneous bands

For procedure h , let $s_{p,h} = \text{sd}_b(\Delta_{p,h}^{(b)})$. We set $q_{.95,h}$ to the empirical 95th percentile of $\max_{p=1}^{28} |\Delta_{p,h}^{(b)} - \bar{\Delta}_{p,h}|/s_{p,h}$ and report $\hat{\Delta}_p \pm q_{.95,h} s_{p,h}$ [31]. The quantile centers deviations on the bootstrap mean, whereas the bands center on the original estimate; band inclusion is not an empirical reversal probability. The maximum includes all 28 primary contrasts, including when reporting within-cohort counts; Arena and SpoofCeleb use their own complete families. No population error-control guarantee is asserted. The historical independent- and shared-trial test reconstructions both separate five of six organizer pairs, without identifying the variant used in the published matrix (S2a).

3.4. Alternative fixed weighting rules

Within each class, every named hierarchy level receives equal conditional mass and trials divide their terminal-group mass equally. Bona-fide rules are empirical trial weights, equal mass over the three sources (ASVspoof, VCC2018, VCC2020), and equal source $\rightarrow$ speaker mass. Spoof rules are empirical trial weights, equal mass over the five spoof strata (ASVspoof, VCC2018 HUB, VCC2018 SPO, VCC2020 Task 1, VCC2020 Task 2), equal mass over observed speaker $\times$ attack cells within stratum, and equal mass successively over stratum, attack family (the attack-type field of the official 21DF trial metadata), attack and speaker. Their 3×4 crossing gives 12 positive, class-normalized rules fixed without inspecting scores. For endpoint w^1 , the path from empirical weights is $w(\lambda) = (1 - \lambda)w^0 + \lambda w^1$, $0 \leq \lambda \leq 1$. Distance is maximum classwise total variation (S5).

4. EXPERIMENTS

Setup. The primary set contains XLSR-Mamba [32], XLSR+SLS (SLS below) [33], XLSR-Conformer [34] and SSL-AASIST [35], plus RawNet2, LFCC-LCNN, LFCC-GMM and CQCC-GMM. All 533,928 evaluation-phase trials have every score. The matched trial/PW comparison uses 5,000 replicates and seed 2026081604; seeds of the other analyses accompany their released outputs.

Primary 21DF result. The matched all-28-pair bootstrap refits the same weighted EER in every replicate. Trial and joint bands separate 26/28 and 18/28 pairs (Table 1); within

Table 1. Top: primary 21DF point EER (%); $n = 533, 928$, empirical class-normalized trial weights. Middle: paired bands (points); primary 21DF uses a 28-pair family, Arena a separate 55-pair family on the same trials. Bottom: separated pairs; primary subsets retain the 28-pair correction and all four arms retain 16/16 cross-cohort separations. SpoofCeleb uses AASIST, SLS, SSL-AASIST and Mamba, its own six-pair family and pooled trial law (stratified check: S8). *One-factor status: Method. †Recorded-seed attack counts: four of five fresh seeds give 21/28 and 2/6 organizer separations (S4b).

SSL detector	EER	Organizer baseline	EER	
XLSR-Mamba	1.884	RawNet2	22.383	
XLS-R+SLS	1.916	LFCC-LCNN	23.477	
XLSR-Conformer	2.273	LFCC-GMM	25.247	
SSL-AASIST	2.852	CQCC-GMM	25.563	
XLSR-Mamba – XLSR-Conformer: -0.389				
Trial $[-0.570, -0.208]$		PW $[-1.348, 0.570]$		
Arena HuBERT-ECAPA – WavLM-ECAPA: -2.153				
Trial $[-2.761, -1.544]$		PW $[-5.241, 0.936]$		
Roster / subset	Trial	Spk.*	Atk.*	Joint
21DF, all	26/28	18/28	22/28†	18/28
21DF, SSL	5/6	2/6	3/6	2/6
21DF, organizer	5/6	0/6	3/6†	0/6
SpoofCeleb	6/6	5/6	3/6	3/6

SSL, the counts fall from 5/6 to 2/6. Mamba–Conformer remains at -0.389 points: its trial band is $[-0.570, -0.208]$, its joint band $[-1.348, 0.570]$. Speaker-only reproduces all eight joint losses; attack-only reproduces four or five of the eight joint losses across Monte Carlo repeats (S4b). All primary within-SSL losses involve Conformer.

Arena comparison. On the eleven complete Speech DF Arena score files for the same 533,928 trials, trial and PW bands separate 52/55 and 38/55 pairs. For HuBERT-ECAPA versus WavLM-ECAPA, the -2.153 -point gap has bands $[-2.761, -1.544]$ and $[-5.241, 0.936]$. A comparison separated under trial perturbations thus loses separation when observed speakers and attacks vary, even at this larger gap.

Pairing and band interpretation. Under the joint law, Mamba and Conformer EERs correlate 0.777; pairing removes about 75% of the sum of their marginal variances, yet the paired SD rises from 0.06149 under trials to 0.31997 points under joint resampling. Table 2 distinguishes mean displacement, sign frequency and band separation. The joint mean displacement is only 0.032 SD, so band centering does not explain inclusion of zero; the trial band excludes zero with 0/5,000 opposite-sign draws, the joint band includes it with 463/5,000 (S4c).

Observed-group influence. The observed speaker group VCC2SM3 supplies 77.09% of Mamba–Conformer’s delete-one-speaker sum of squared deviations and 55.85% for

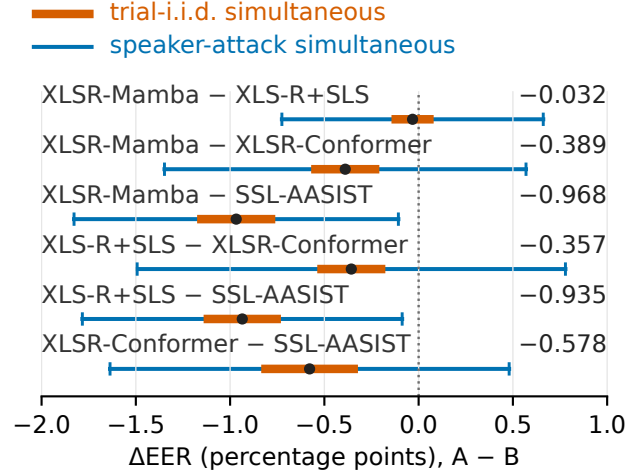


Fig. 1. All six within-SSL pairs on primary 21DF. Dots and right-hand labels are point gaps ($A - B$; negative favors A). Thick orange: trial bands; thin blue with end caps: PW bands. Both retain the full 28-pair simultaneous max- t construction. Five trial bands and two PW bands exclude zero; the point estimates do not change.

Table 2. Mamba minus Conformer: mean shift from the point gap, pair SD (points), opposite-sign draws out of 5,000, and zero exclusion under the full 28-pair band.

Law	Mean shift	SD	Opposite	Excludes?
Trial	-0.002492	0.061488	0	Yes
Speaker	-0.003175	0.270826	210	No
Attack	-0.010235	0.147333	37	No
Joint	-0.010135	0.319971	463	No

SLS–Conformer. Deleting that group moves the Mamba–Conformer gap from -0.389 to -0.129 points, locating much of the sensitivity in this observed group (S4e).

Composition-preserving control. Using 1,000 retained replicates, a constrained PW arm resamples within each bona-fide source or spoof stratum, conditions on retained class support, and restores each source and stratum mass to its observed value; it also separates 18/28 and 0/6 organizer pairs. This arm was designed after the main result and is a robustness check, not prospective evidence (S6). Thus neither threshold handling nor movement of these eight class-specific masses is necessary for the contrast.

Check on single-source SpoofCeleb. On all 91,130 evaluation trials [36], pooled trial resampling separates 6/6 pairs (a class-stratified recomputation also separates 6/6) and joint resampling 3/6. This is an off-domain sensitivity check on a different roster. The SpoofCeleb EERs are 57.93%, 24.51%, 26.72%, and 27.58% for AASIST, SLS, SSL-AASIST, and Mamba. Joint resampling retains only the three comparisons with AASIST. Every base item has one bona-fide and nine

spoof renditions under one source label, so between-source mass cannot move; speaker and spoof-attack multiplicities still vary. Attack-only matches the joint indicator vector, whereas speaker-only retains 5/6. The three detectors shared with primary 21DF have 2/3 separated in every primary arm under the original 28-pair correction (S4c); the contrasting factor patterns therefore depend on the score collections. Access chronology and scoring provenance are in S8.

ASVspoof 5 check. An EER-only four-detector check also changes separation from 6/6 under trial resampling to 5/6; the SSL-AASIST and AASIST score files lack their originating run logs (S8).

Alternative weights. Fixed rules reverse 5/6 organizer-baseline, 1/6 within-SSL and 0/16 cross-cohort point orderings. The SSL pair is Mamba-SLS (gap -0.032 points), already unseparated under trial resampling. Recomputing both bootstraps with each rule’s weights as base weights (computed after the primary results, 5,000 draws per rule and bootstrap) separates all 16 cross-cohort pairs in all 24 rule $\times$ law cells; the closest band endpoint to zero is -12.79 points. S5 gives a constructive Mamba-SLS reversal and its searched masses. The reversing sign also holds at distinct-score boundaries (supplement, Sec. S5).

Surviving modern comparisons. Of the two within-SSL pairs separated by joint resampling, Mamba-SSL-AASIST loses separation when VCC2018 is omitted (gap $-0.968 \rightarrow -0.270$ points); SLS-SSL-AASIST loses it when VCC2020 is omitted ($-0.935 \rightarrow -0.760$). For source deletion we use our jackknife sensitivity construction defined in S2, with the contrast variance floored at the larger one-factor variance and Gaussian max- t calibration over all 28 pairs. Related to max-se methods [37], it preserves all 16 cross-cohort separations without a coverage guarantee. The first loss combines gap shrinkage with the changed band; the second retains a sizeable gap but gains uncertainty.

5. DISCUSSION

Reporting decision rule. Choose the perturbation to match the reporting question: individual trials, or weights of observed speakers and attacks. Report the point gap and both bands when a comparison is intended to withstand both checks. If the separation indicators disagree, label the comparison procedure-sensitive within the full comparison family and do not present trial-only separation as evidence that it survives the group check. Publish trial-to-speaker/attack membership and source weights alongside these results.

Coverage boundary. Coverage fell below .90 in 16/24 low-EER cells for each jackknife interval and 11/24 for PW percentile; all three failed in the same 11 cells (S7). These imposed-model simulations evaluate different intervals from the principal all-pair bands and do not establish a population sampling model for 21DF.

Artifact. Supplement S4a–S4e supplies the complete

four-arm bands, Monte Carlo repeats, mean/sign and pairing diagnostics, weighted-tie checks and group influence. The three evidence JSON files and analysis code accompany artifact version

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at github.com/rvirgilli/asvspoof2021-df-clustered-uncertainty; S8 gives file selectors and hashes.

6. CONCLUSION

Trial-resampling separation disappears under speaker-attack perturbations for Arena HuBERT-WavLM and primary Mamba-Conformer; for the latter, this occurs despite 75% marginal-variance cancellation from pairing. Speaker-only matches the joint indicators on primary 21DF; attack-only does so on the separate SpoofCeleb roster, where trial-to-joint separation falls from 6/6 to 3/6. Cross-cohort separation survives the tested laws, weights and source deletions, while modern within-cohort comparisons expose concentrated group sensitivity.

7. ACKNOWLEDGMENT

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8. COMPLIANCE WITH ETHICAL STANDARDS

This study analyzed previously collected ASVspoof 2021, ASVspoof 5 and SpoofCeleb recordings and detector scores. It involved no new recording or interaction with human participants; no ethical approval was required. The authors declare no competing interests.

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